Customer Analytics EDA Mini Project Documentation

Overview

This project performs Exploratory Data Analysis (EDA) on a customer analytics dataset to understand purchasing behavior and demographic patterns. The analysis helps in identifying key trends, correlations, and insights that can inform business decisions.

Project Workflow

1. Data Inspection - Loading and initial examination of the dataset.
2. Data Cleaning - Handling missing values and removing duplicates.
3. Univariate Analysis - Analyzing individual variables.
4. Bivariate Analysis - Exploring relationships between two variables.
5. Correlation Analysis - Examining correlations between numerical variables.
6. Insight Generation - Drawing conclusions from the analysis.

Technologies Used

- Python: Programming language for data analysis
- Pandas: Data manipulation and analysis
- NumPy: Numerical computing
- Matplotlib: Plotting library
- Seaborn: Statistical data visualization
- Jupyter Notebook: Interactive computing environment

Dataset Introduction

This dataset contains information about customer demographics and purchasing behavior collected from a retail business. The data is used to understand how different customer characteristics influence purchasing patterns.
Each row in the dataset represents one individual customer transaction or customer record. It provides details about the customer's personal and financial attributes along with their purchasing activity.
The columns describe different attributes such as:
- CustomerID – Unique identifier for each customer
- Age – The age of the customer
- Gender – The gender category of the customer
- City – The city of residence
- Education – Education level
- MaritalStatus – Marital status
- AnnualIncome – The annual income level of the customer
- SpendingScore – Customer spending score
- YearsEmployed – Years of employment
- PurchaseFrequency – Frequency of purchases
- OnlineVisitsPerMonth – Monthly online visits
- ReturnedItems – Number of returned items
- PreferredDevice – Preferred shopping device
- LastPurchaseAmount – Amount of last purchase

The objective of this analysis is to explore patterns in customer behavior, understand how individual variables are distributed, and identify relationships between demographic factors and purchasing activity using Exploratory Data Analysis (EDA). The insights obtained from this analysis can help businesses better understand their customers and make data-driven decisions.

Data Cleaning

Data Cleaning Decisions

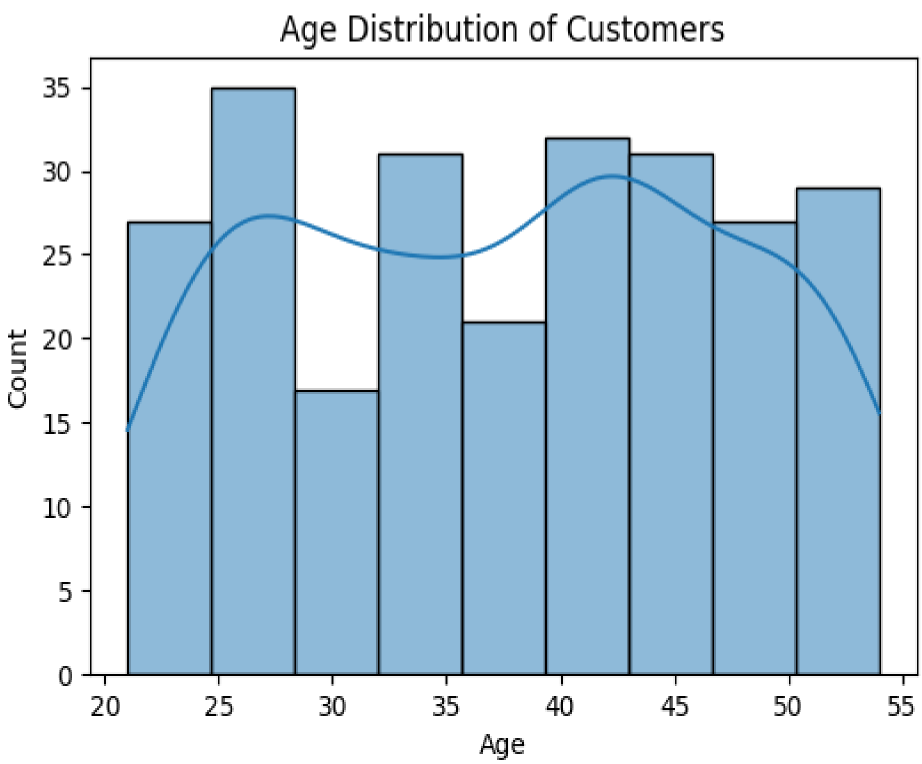
- The 'Education' column contained 12 missing values. Since it is a categorical feature, missing values were replaced using the mode (most frequent category).
- The 'AnnualIncome' column also contained 12 missing values. These were filled using the median value because income data may contain extreme outliers.
- Five duplicate rows were identified and removed to prevent biased analysis results.

After preprocessing, the dataset became clean and suitable for exploratory analysis.

Univariate Analysis

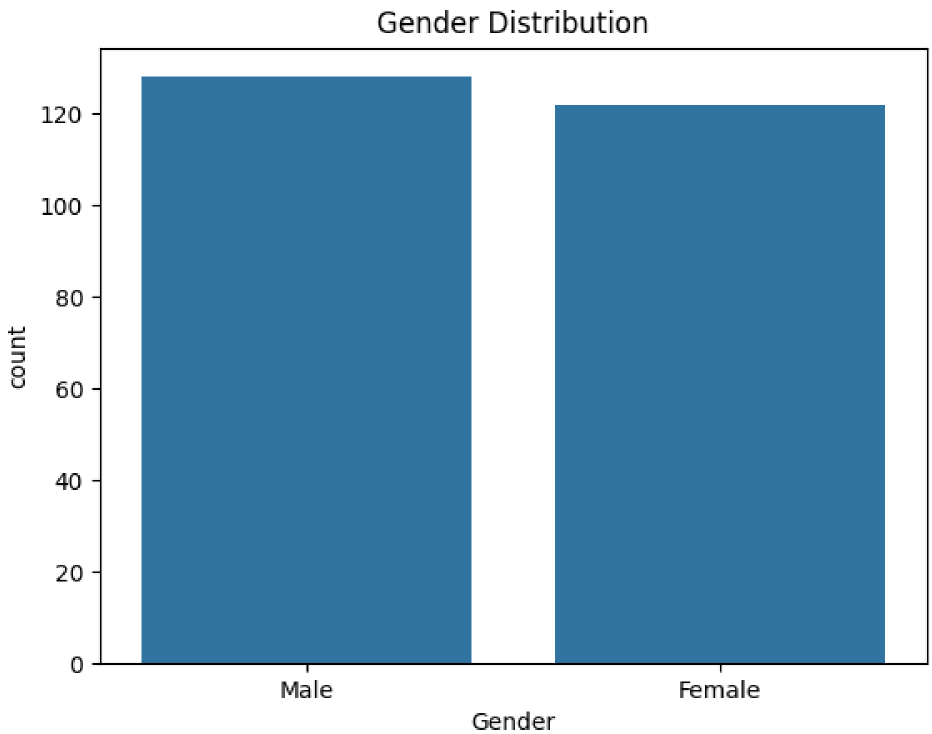
Age Distribution

Most customers fall between ages 30 and 45, indicating that the business mainly attracts middle-aged individuals.



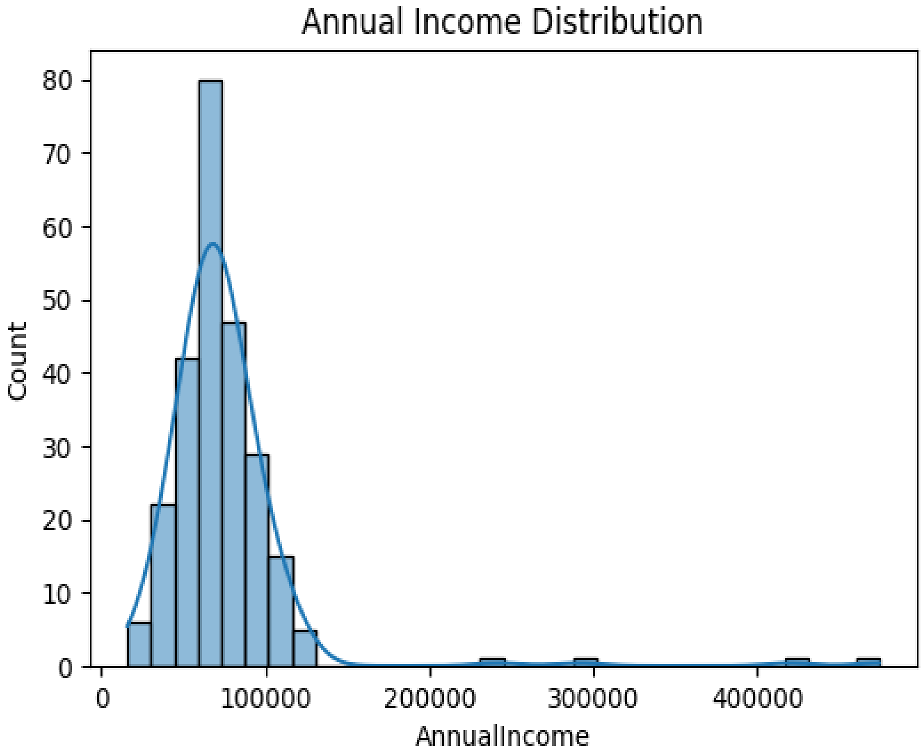
Gender Distribution

The dataset shows a relatively balanced gender distribution, allowing unbiased comparison between male and female customers.



Annual Income Distribution

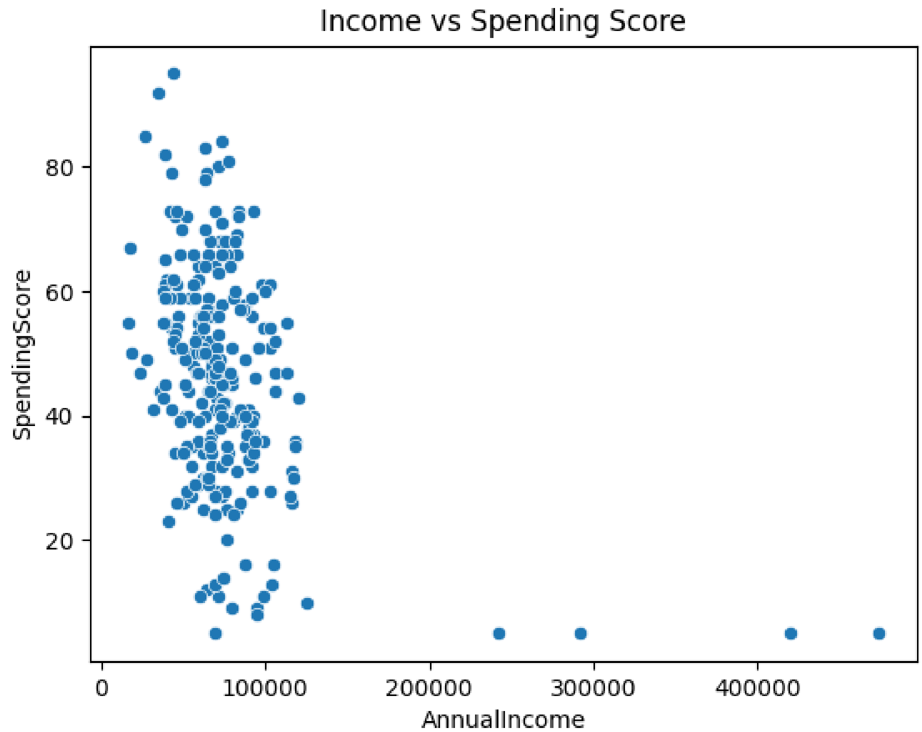
Income distribution is slightly right-skewed, meaning most customers fall into moderate income ranges while a few earn significantly higher incomes.



Bivariate Analysis

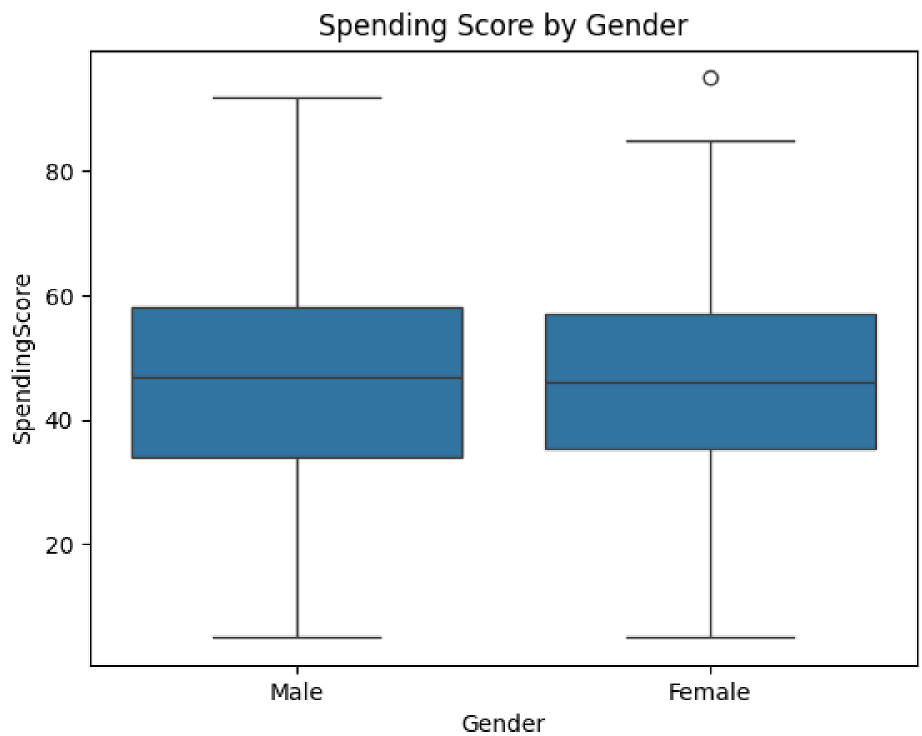
Income vs Spending Score

The scatter plot shows that higher income does not always result in higher spending, suggesting different customer spending behaviors across income groups.



Spending Score by Gender

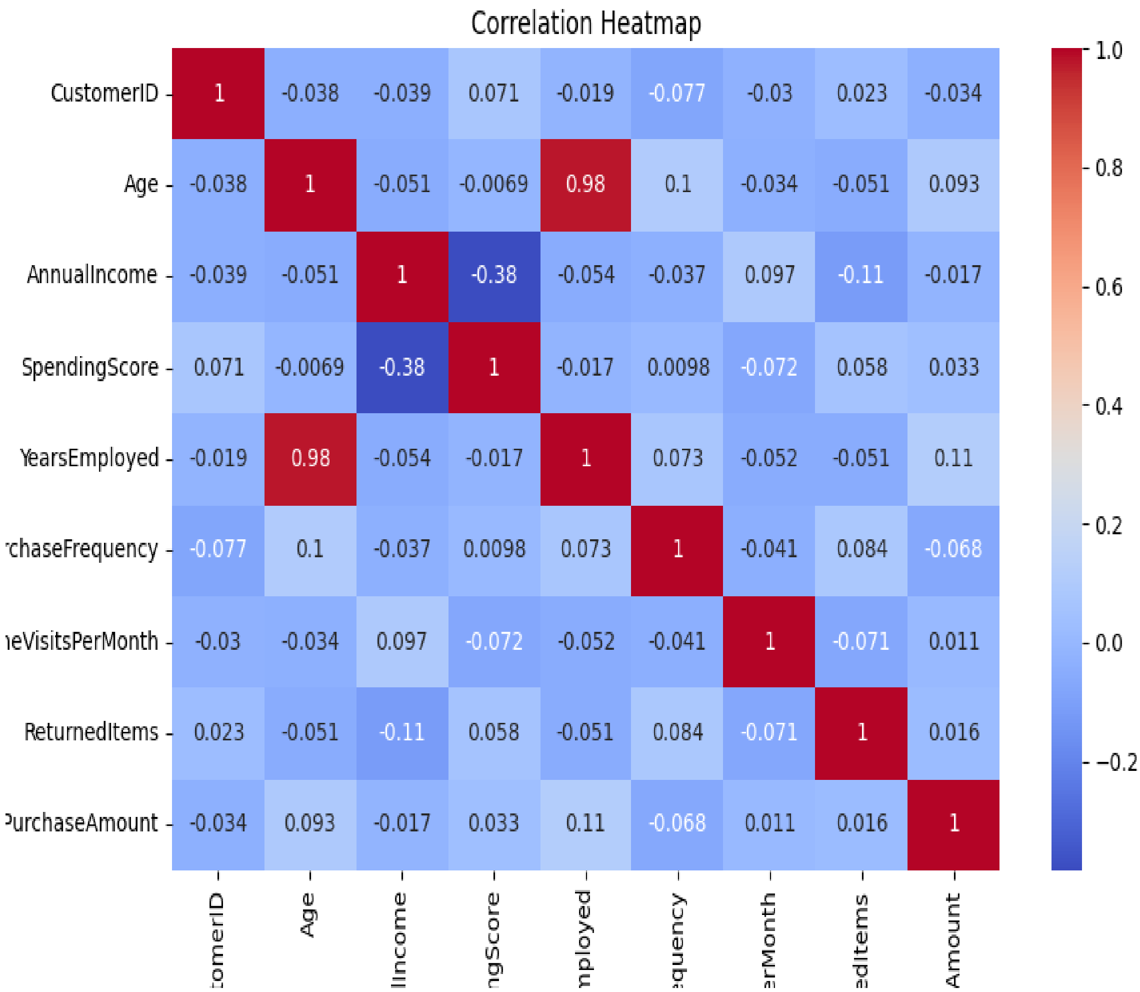
The boxplot reveals variation in spending behavior between genders, with both groups showing a wide range of spending scores.



Multivariate Analysis

Correlation Heatmap

The correlation heatmap shows relationships between numerical variables. Strong correlations help identify factors that influence customer spending and purchasing behavior.



Key Insights

1. Most customers belong to the middle-age group (30–45 years), indicating the primary target demographic.
2. Annual income does not strongly determine spending behavior, suggesting psychological or lifestyle factors influence purchases.
3. Purchase-related variables such as spending score and purchase frequency show moderate relationships, indicating consistent buying patterns among active customers.
4. Customer spending varies independently of income.
5. Middle-aged customers form the majority group.
6. Clean data significantly improves analysis reliability.

Conclusion

Overall, the dataset provides valuable insights into customer demographics and purchasing behavior, which can help businesses improve marketing strategies and customer segmentation. The analysis reveals that while demographic factors like age and gender play a role in customer behavior, income alone is not a strong predictor of spending. Further analysis could include predictive modeling to forecast customer behavior based on these insights.

How to Run the Project

Install dependencies:
pip install -r requirements.txt
Run notebook:
jupyter notebook MiniProject1_EDA.ipynb